

PS 2 Discussion :

$$15) \quad P(X = x) \geq P(X = x-1)$$

Let's say PMF of X is non-decreasing till this value of x .

$${}^n C_x p^x (1-p)^{n-x} \geq {}^n C_{x-1} p^{x-1} (1-p)^{n-x+1}$$

$$\frac{\cancel{n!}}{x! (n-x)!} p^x (1-p)^{n-x} \geq \frac{\cancel{n!}}{(x-1)! (n-x+1)!} p^{x-1} (1-p)^{n-x+1}$$

$$\frac{p}{x} \geq \frac{1-p}{n-x+1}$$

$$np - \cancel{px} + p \geq x - \cancel{x}$$

$$\boxed{x \leq p(n+1)}$$

III) for $Y \sim \text{Poisson}(\lambda)$, Y be non-decreasing till $Y = y$

$$\Rightarrow P(Y = y) \geq P(Y = y-1)$$

$$\frac{\cancel{e^{-\lambda}} \lambda^y}{y!} \geq \frac{\cancel{e^{-\lambda}} \lambda^{y-1}}{(y-1)!}$$

$$\lambda \geq y \Rightarrow \underline{y \leq \lambda}$$

$$6) P(S=s | A=a) = \frac{P(\{S=s\} \cap \{A=a\})}{P(A=a)}$$

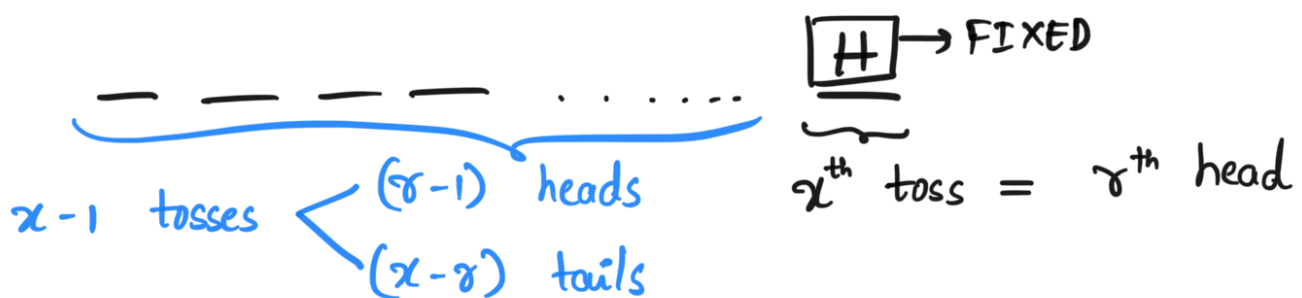
$\underbrace{\{S=s\} \cap \{A=a\}}_{\text{Let 'E'}} \rightarrow \begin{array}{l} \text{'s' couples alive \& a} \\ \text{total of 'a' people} \\ \text{alive} \end{array}$

$$P(E) = {}^m C_s p^{2s} \underbrace{{}^{m-s} C_{a-2s}}_{\text{Selecting } a-2s \text{ 'COUPLES' (not individual people) where only 1 partner is alive}} \underbrace{(p(1-p) + (1-p)p)^{a-2s}}_{P(1 \text{ partner alive}) = 2p(1-p)} \underbrace{(1-p)^{2m-a}}_{\substack{\text{The rest} \\ (2m-a) \\ \text{'PEOPLE' } \\ \text{dead.}}}$$

prob. that both partners of 's' couples are alive

11. (b). ii.

$X \sim$ No. of tosses until exactly 'r' heads seen.



$$P_X(x) = (p)^r x^{x-1} C_{x-1}^{r-1} p^{r-1} (1-p)^{x-r}$$

$$P_X(x) = {}^{x-1}C_{r-1} p^r (1-p)^{x-r} \quad (x \geq r)$$

$$E[X] = r \sum_{x=r}^{\infty} \frac{x}{{}^{x-1}C_{r-1}} p^r (1-p)^{x-r}$$

$$= r \sum_{x=r}^{\infty} x {}^{x-1}C_{r-1} p^r (1-p)^{x-r}$$

$$= r p^r \sum_{x=r}^{\infty} x {}^{x-1}C_{r-1} (1-p)^{x-r}$$

$$= r p^r \left[{}^rC_{r-1} (1-p)^0 + {}^{r+1}C_{r-1} (1-p)^1 + \dots \right]$$

$$= r p^r \left[1 + {}^{r+1}C_1 (1-p) + {}^{r+2}C_2 (1-p)^2 + \dots \right]$$

$$** (1-x)^{-n} = 1 + {}^nC_1 x + {}^{n+1}C_2 x^2 + {}^{n+2}C_3 x^3 + \dots$$

$$\left(\text{If } x = (1-p) \text{ \& } n = r+1 \right)$$

$$(p)^{-r-1} = 1 + {}^{r+1}C_1 (1-p) + {}^{r+2}C_2 (1-p)^2 + \dots$$

$$\therefore E[X] = r p^r \times (p)^{-r-1}$$

$$\Rightarrow \boxed{E[X] = \frac{r}{p}}$$

* Another approach

$$\text{We know } P_X(x) = {}^{x-1}C_{r-1} p^r (1-p)^{x-r} ; (x \geq r)$$

$$\& \sum_{x=r}^{\infty} P_X(x) = 1 \quad (\text{Property of PMF})$$

$$p \sum_{x=r}^{\infty} \frac{x-1}{p} C_{x-1} p^r (1-p)^{x-r} = 1$$

$$\Rightarrow \underbrace{\sum_{x=r}^{\infty} (x-1) C_{x-1} p^{r-1} (1-p)^{x-r}}_{\text{Same as } E[X] \times (1/p)} = 1/p$$

$$\therefore E[X] = \underline{\underline{r/p}}$$

3) (a) The number of ways in which 8 people left the elevator across 6 floors can be modelled as:

$$f_1 + f_2 + f_3 + f_4 + f_5 + f_6 = 8$$

where f_i = no. of people who left in the i^{th} floor.

Since the elevator went until 6th floor (and got empty), at least 1 person must have left in the 6th floor $\Rightarrow f_6 \geq 1$

$\therefore f_6$ is atleast 1, we can replace f_6

with f_6' in eq. (1) & make the RHS 7.

$$\Rightarrow f_1 + f_2 + f_3 + \dots + f_6' = 7$$

Can be thought of as distributing/grouping 7 identical items among/into 6 different groups.



One example of grouping $\Rightarrow$ We can thus find total no. of way by just arranging 7 (0)'s & 5 (1)'s

$$\Rightarrow \frac{12!}{7! \times 5!}$$

(b) A similar approach as (a) can be followed but separately for ⁽⁵⁾men & ₍₃₎women. All ways

women, and the total no. of
can be found by multiplying the
two.